

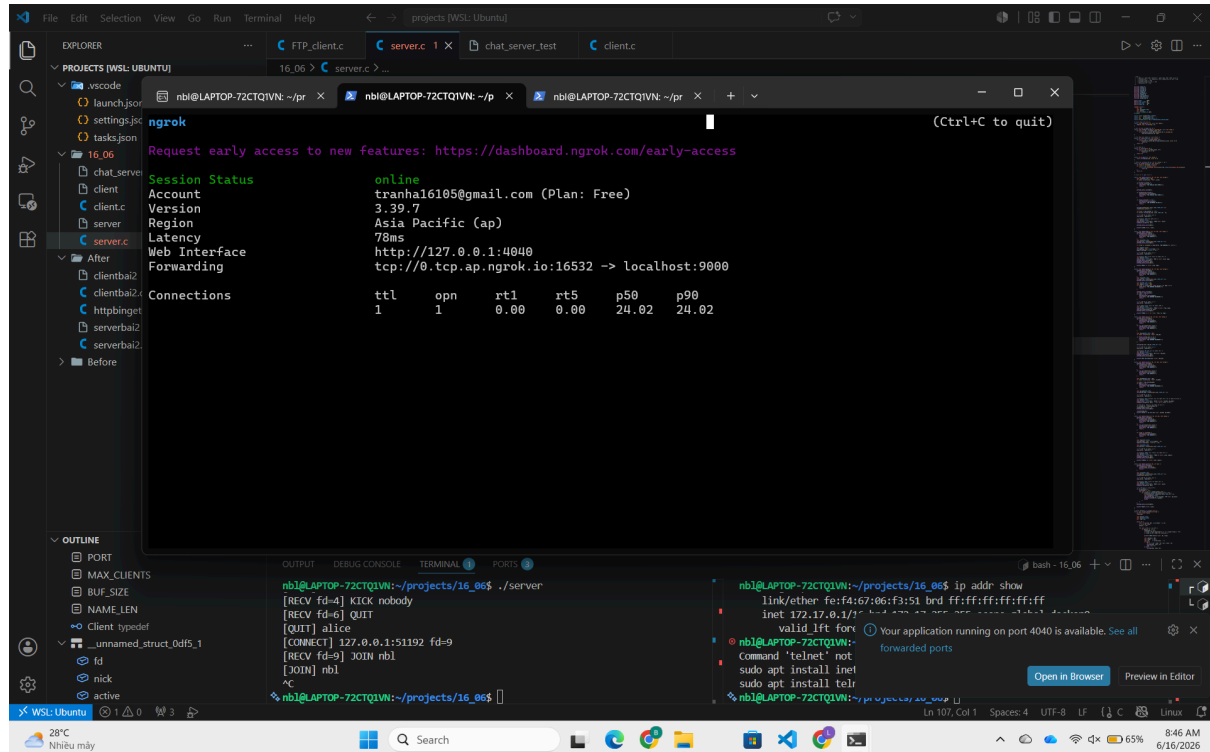
Bài tập nhóm 16/06/2026

Học phần: Lập trình mạng

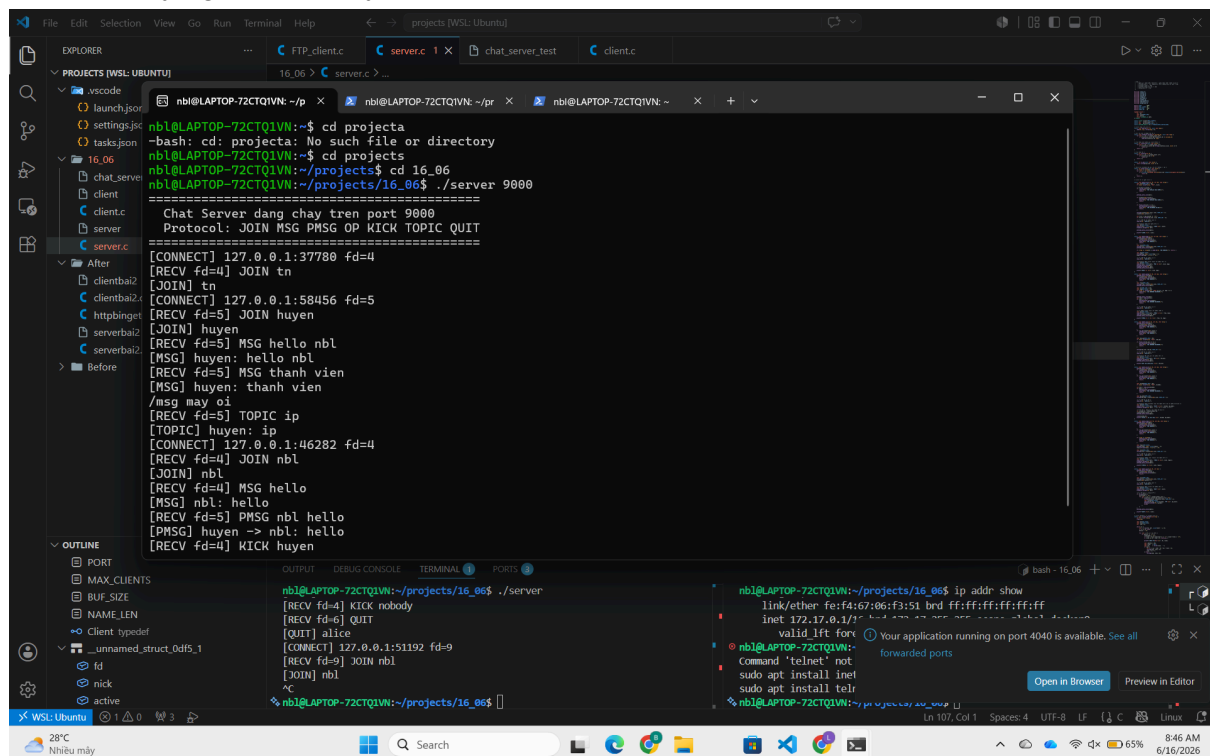
Mã lớp: 168503

Sinh viên làm ứng dụng chat_server.c: Nguyễn Bảo Linh - MSSV: 20235362

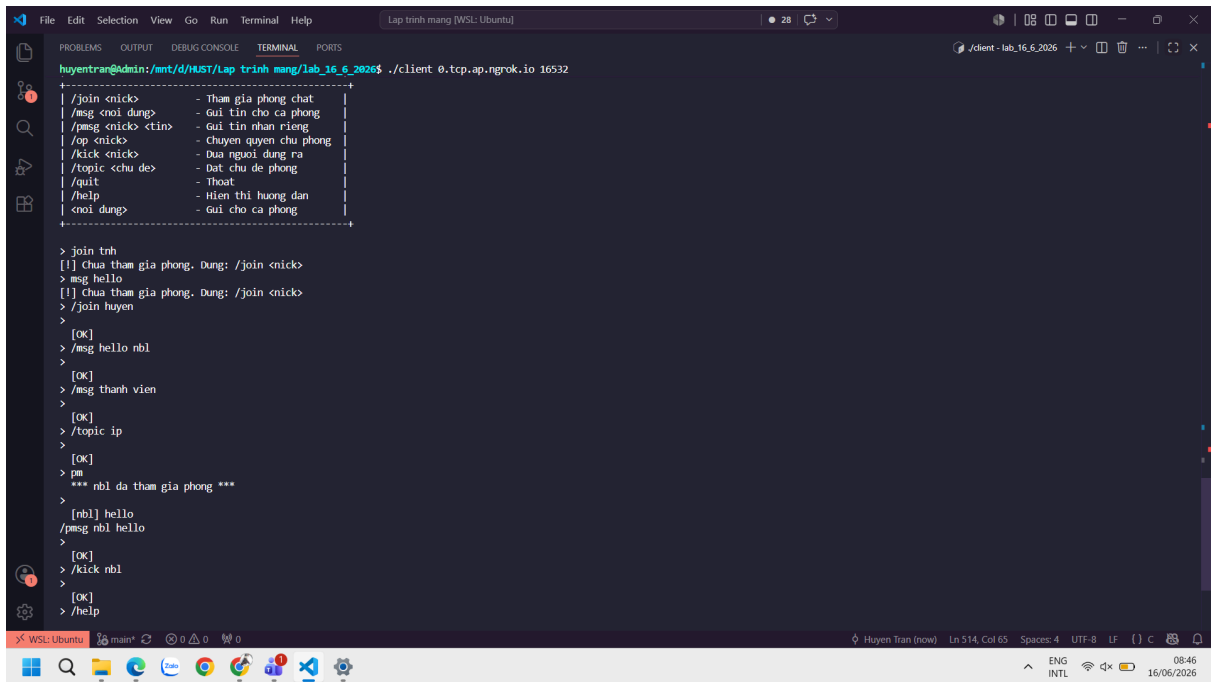
Sinh viên làm ứng dụng chat_client.c: Trần Ngọc Huyền - MSSV: 20235347



Hình 1. Chạy ngrok trên máy Server



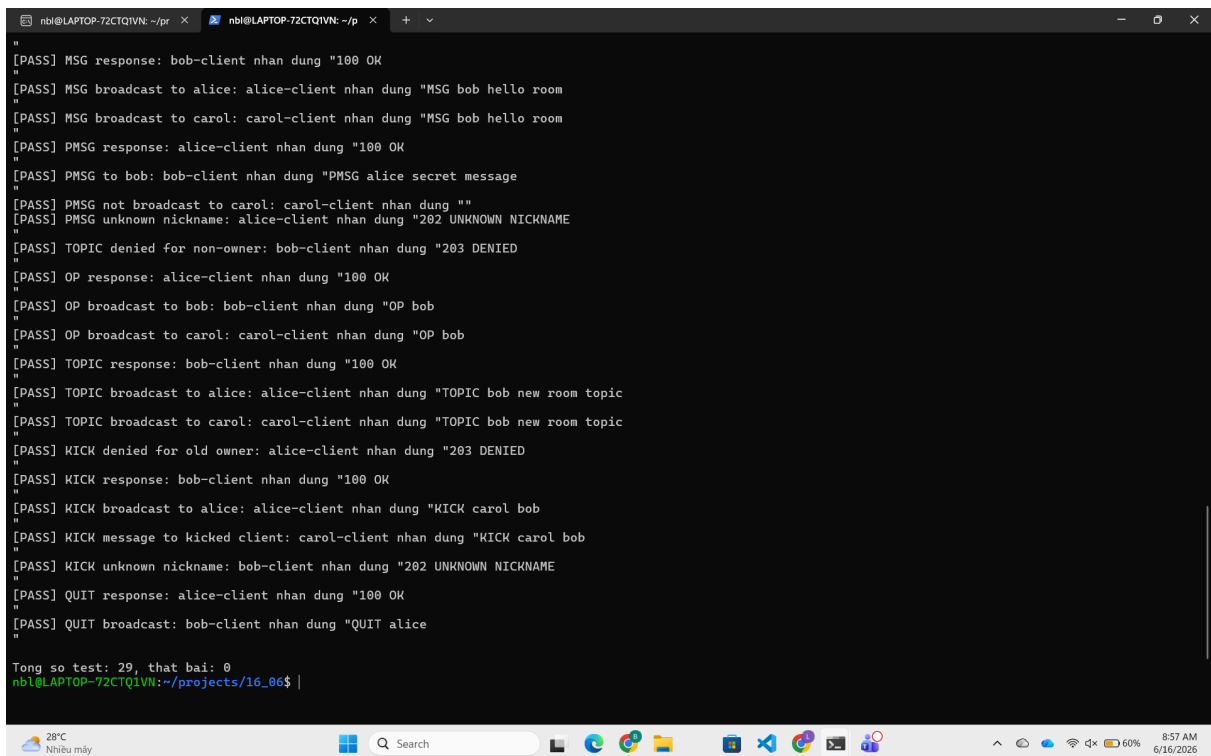
Hình 2. Chạy Server trên máy Server



```
huyentr@Admin:~/mnt/d/HUST/Lap_trinh_mang/lab_16_e_2026$ ./client 0.tcp.ap.ngrok.io 16532
-----
| /join <nick>      - Tham gia phong chat |
| /msg <noi dung>   - Gui tin cho ca phong |
| /msg <nick> <tin> - Gui tin nhan rieng |
| /op <nick>        - Chuyen quyen chu phong |
| /kick <nick>      - Dua nguoi dung ra |
| /topic <chu de>   - Dat chu de phong |
| /quit            - Thoat |
| /help            - Hien thi huong dan |
| <noi dung>       - Gui cho ca phong |
-----

> join trn
[!] chua tham gia phong. Dung: /join <nick>
> msg hello
[!] chua tham gia phong. Dung: /join <nick>
> /join huyen
>
[OK]
> /msg hello nbl
>
[OK]
> /msg thanh vien
>
[OK]
> /topic ip
>
[OK]
> pa
*** nbl da tham gia phong ***
>
[nbl] hello
/msg nbl hello
>
[OK]
> /kick nbl
>
[OK]
> /help
```

Hình 3. Chạy ứng dụng client trên máy thành viên lập trình chat_client.c



```
nb1@LAPTOP-7ZCTQ1VN: ~/pr x nb1@LAPTOP-7ZCTQ1VN: ~/p x + v
"
[PASS] MSG response: bob-client nhan dung "100 OK
"
[PASS] MSG broadcast to alice: alice-client nhan dung "MSG bob hello room
"
[PASS] MSG broadcast to carol: carol-client nhan dung "MSG bob hello room
"
[PASS] PMSG response: alice-client nhan dung "100 OK
[PASS] PMSG to bob: bob-client nhan dung "PMSG alice secret message
"
[PASS] PMSG not broadcast to carol: carol-client nhan dung ""
[PASS] PMSG unknown nickname: alice-client nhan dung "202 UNKNOWN NICKNAME
"
[PASS] TOPIC denied for non-owner: bob-client nhan dung "203 DENIED
[PASS] OP response: alice-client nhan dung "100 OK
[PASS] OP broadcast to bob: bob-client nhan dung "OP bob
[PASS] OP broadcast to carol: carol-client nhan dung "OP bob
[PASS] TOPIC response: bob-client nhan dung "100 OK
[PASS] TOPIC broadcast to alice: alice-client nhan dung "TOPIC bob new room topic
[PASS] TOPIC broadcast to carol: carol-client nhan dung "TOPIC bob new room topic
[PASS] KICK denied for old owner: alice-client nhan dung "203 DENIED
[PASS] KICK response: bob-client nhan dung "100 OK
[PASS] KICK broadcast to alice: alice-client nhan dung "KICK carol bob
[PASS] KICK message to kicked client: carol-client nhan dung "KICK carol bob
[PASS] KICK unknown nickname: bob-client nhan dung "202 UNKNOWN NICKNAME
[PASS] QUIT response: alice-client nhan dung "100 OK
[PASS] QUIT broadcast: bob-client nhan dung "QUIT alice
"
Tong so test: 29, that bai: 0
nb1@LAPTOP-7ZCTQ1VN:~/projects/16_06$ |
```

Hình 4. Chạy chương trình test chat_server_test được cung cấp